

less of environmental conditions.

In the Asian market, companies like Hyundai and Toyota have taken an interest in the field. Even worldwide, manufacturers like Audi and BMW are already working on hydrogen vehicles. Numerous experts are expecting a lot from this alternative, while others still remain skeptical.

Hydrogen-powered vehicles require high-pressure compression to work, which is difficult to achieve due to the low density of hydrogen. Also, there are a limited number of refueling stations that exist in just a few countries (like Germany and the US) due to lack of sufficient manufacturing and demand.

The low demand for hydrogen vehicles stems from the fact that these are quite expensive to buy as well as use. The overall efficiency is almost half of what BEV has due to losses in hydrogen production through electrolysis.

New techniques can help in alleviating risks regarding the safety of such a vehicle. For instance, researchers at Northwestern University in the US, in collaboration with the Colorado School of Mines and National Institute for Standards and Technology, have recently come up with novel adsorbent materials that are built from organic molecules and metal ions or clusters and can store hydrogen at

much lower pressures.

It is also crucial to develop advanced infrastructure to refuel these vehicles and shorten transportation distances and make policies to increase their adoption. In the US, the California Energy Commission (CEC) and the state's Air Resources Board provide subsidies for new hydrogen car buyers. South Korea plans to expand the country's hydrogen refuelling infrastructure to 660 pumps by 2030. Completely switching to such clean energies will go a long way in keeping the air clean and ensuring sustainable mobility.

—Ayushee Sharma

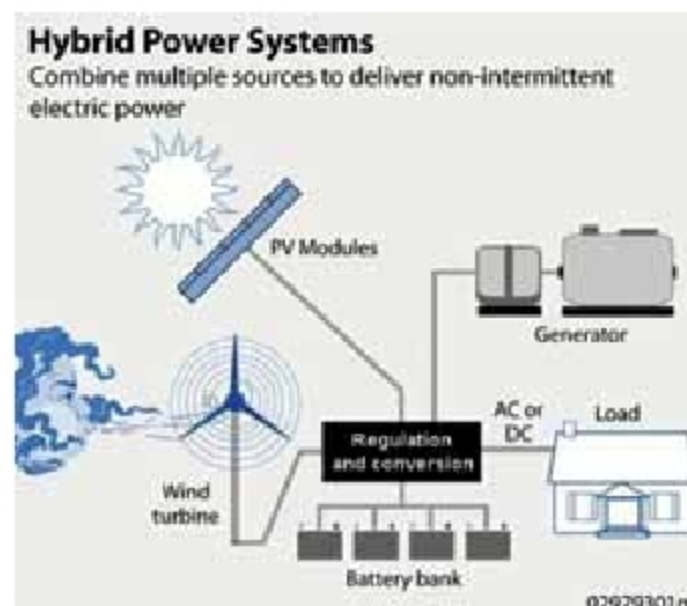
2 WIND-SOLAR HYBRID SYSTEMS TO IMPROVE GRID STABILITY

Using renewable energy sources like wind and solar for power generation has helped in reducing carbon footprints and made power available to non-electrified remote areas. A hybrid power system that combines the benefits of both sources can improve the reliability and stability of the grid to deliver non-intermittent electric power

Sustainable power generation through renewable energy, instead of conventional fossil fuels, has not only helped in reducing carbon footprints but also made power available to non-electrified remote areas. While choosing green energy options for a region, several factors like geographic variability, economic feasibility, and so on need to be considered.

For complete reliance on renewable energy like solar or wind, it must be available in all seasons. However, this is not the actual case due to unpredictable weather. For example, solar energy cannot be harnessed to its full potential in monsoon, and wind speeds may be highest in one season and not so in others. In areas where more than one renewable energy like wind or solar is available, a hybrid power system that combines their benefits can improve the reliability and stability of the grid.

Wind-solar hybrid systems interface wind turbine generators and solar panels to generate power in a decentralised mode, typically in the range of 1kW to 10kW. The standalone system is connected to a charge controller that



can handle both systems for charging the battery so that energy can be used whenever needed. A compatible inverter converts DC output to AC for purposes like power distribution, powering appliances like refrigerator and oven, and so on. In case of on-grid hybrid systems, the battery is charged by the grid when both solar and wind are not available.

These systems can be installed on rooftops if size permits. The cost varies based on the ratio of wind and solar components and maybe even cheaper than a solar system. The size

of the battery storage also gets reduced, which makes it easier to handle.

The generation of power from wind energy is done by wind turbines (commonly vertical or horizontal axis), whose blades rotate with wind blowing within the range of defined cut in (usually, a minimum of 4-5m/s) and cut off speeds, coupled with generators. The axis of rotation of the turbines must align with the wind direction.

Solar power systems have solar panels, which consist of photovoltaic (PV) cells that utilise sunlight to generate electrical energy. Factors like sunlight intensity and duration, as well as cleanliness of panels and their orientation, need to be considered for efficient solar panel arrangement. Some arrangements also involve moving solar panels in an accurate direction by tracking the sun to get more light.

Hybrid systems are often used for small loads. In the residential sector, there has been significant adoption—particularly in single-family homes, luxury apartments, and large multi-